

**FACULTY OF SCIENCE AND TECHNOLOGY**

**LOGIC BUILDING PROGRAMMING & VERSION CONTROL REPORT**

**PROGRAMME:** BSSE/BSCS/BIT

**YEAR/SEM:** YEAR 2 SEM 1

**COURSE CODE:** BIT2115

**PROBLEM NAME:** JAVA ACCESS MODIFIERS

**STUDENT DETAILS:**

|                        |                  |
|------------------------|------------------|
| <b>NAME OF STUDENT</b> | Dureti Edao Obse |
| <b>REG NUMBER</b>      | 25/1707/BSCS-S   |



**FACULTY OF SCIENCE AND TECHNOLOGY**  
**LOGIC BUILDING PROGRAMMING & VERSION CONTROL REPORT**

**Table of Contents**

**FACULTY OF SCIENCE AND TECHNOLOGY**

**LOGIC BUILDING PROGRAMMING & VERSION CONTROL REPORT**

**List of Figures**

|                                                    |   |
|----------------------------------------------------|---|
| Figure 1:Class Main showing project structure..... | 4 |
| Figure 2:Output after running the program .....    | 5 |

**FACULTY OF SCIENCE AND TECHNOLOGY**

**LOGIC BUILDING PROGRAMMING & VERSION CONTROL REPORT**

**Section 1**

**Logic Building & Programming**

**Lines of code & explanations**

| <b>Line Number</b>        | <b>Explanation</b>                                                                                                                                                                                                |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Patient.java: 1           | Places the class in the package hospital.records, which holds the patient data classes.                                                                                                                           |
| Patient.java: 3           | Declares the public class Patient. It is public so classes in other packages (reception, doctors) can use it.                                                                                                     |
| Patient.java: 4-5         | name and appointmentDate are private: the data is hidden and can only be read through the public getters, so no other class can change it directly.                                                               |
| Patient.java: 6           | diagnosis is private because it is sensitive medical information. Only code inside Patient can touch it directly.                                                                                                 |
| Patient.java: 8-10        | Public constructor that sets name, appointment date and diagnosis when a patient object is created.                                                                                                               |
| Patient.java: 11-12       | getName() and getAppointmentDate() are public: receptionists (and anyone) may view these basic details, but they are read-only.                                                                                   |
| Patient.java: 13-14       | getDiagnosis() and setDiagnosis() are protected: only subclasses (such as MedicalFile used by doctors) and the same package can view or update the diagnosis. Receptionists cannot.                               |
| MedicalFile.java: 1       | Places the class in a different package, hospital.medical, to separate medical staff code from patient records.                                                                                                   |
| MedicalFile.java: 3       | Imports Patient from the hospital.records package so it can be extended.                                                                                                                                          |
| MedicalFile.java: 5       | MedicalFile extends Patient. Inheritance is what allows it to use the protected methods from another package.                                                                                                     |
| MedicalFile.java: 6       | Public constructor that passes the patient details up to the Patient class using super().                                                                                                                         |
| MedicalFile.java: 7-8     | viewDiagnosis() and updateDiagnosis() are public methods for doctors. They call the protected getDiagnosis() and setDiagnosis(), so doctors can view and update medical information.                              |
| FinancialRecord.java: 1   | Places the class in the hospital.admin package (the administration department).                                                                                                                                   |
| FinancialRecord.java: 3   | The class has no modifier, so it is package-private: only classes inside hospital.admin can see or create it. This keeps financial records within administration.                                                 |
| FinancialRecord.java: 4-6 | The amountBilled field, the constructor and balanceDue() are also package-private, so they cannot be accessed from doctors' or receptionists' code. balanceDue() returns the amount billed minus the amount paid. |

**FACULTY OF SCIENCE AND TECHNOLOGY**

**LOGIC BUILDING PROGRAMMING & VERSION CONTROL REPORT**

|                               |                                                                                                                                               |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Administration.java:<br>1, 3  | Puts the class in hospital.admin and makes it public, so it is the only door to the financial data for outside code.                          |
| Administration.java:<br>4-6   | showBalance() is public. It creates a FinancialRecord (allowed, same package) and prints the balance due, without exposing the record itself. |
| HospitalSystem.java:<br>1-5   | Declares the package hospital and imports the Administration, MedicalFile and Patient classes from their packages.                            |
| HospitalSystem.java:<br>7-8   | The public class HospitalSystem and its main method, where the program starts.                                                                |
| HospitalSystem.java:<br>9     | Creates a MedicalFile object (the doctor's view of the patient) with a name, appointment date and diagnosis.                                  |
| HospitalSystem.java:<br>10-11 | The receptionist is given only the Patient type, so the receptionist can view only the name and appointment date through the public getters.  |
| HospitalSystem.java:<br>12-13 | Commented lines that show what would NOT compile: getDiagnosis() is protected and diagnosis is private, so a receptionist cannot access them. |
| HospitalSystem.java:<br>14-16 | The doctor views the diagnosis and then updates it using the public methods of MedicalFile, and prints both results.                          |
| HospitalSystem.java:<br>17    | Administration shows the balance due (500000 billed, 200000 paid). Only Administration can work with the financial record.                    |
| HospitalSystem.java:<br>18    | Commented line showing that new FinancialRecord(...) would not compile from here because the class is package-private.                        |
| HospitalSystem.java:<br>19-20 | Closes the main method and the class.                                                                                                         |

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**LOGIC BUILDING PROGRAMMING & VERSION CONTROL REPORT**

**Detailed Screenshots and caption descriptions**

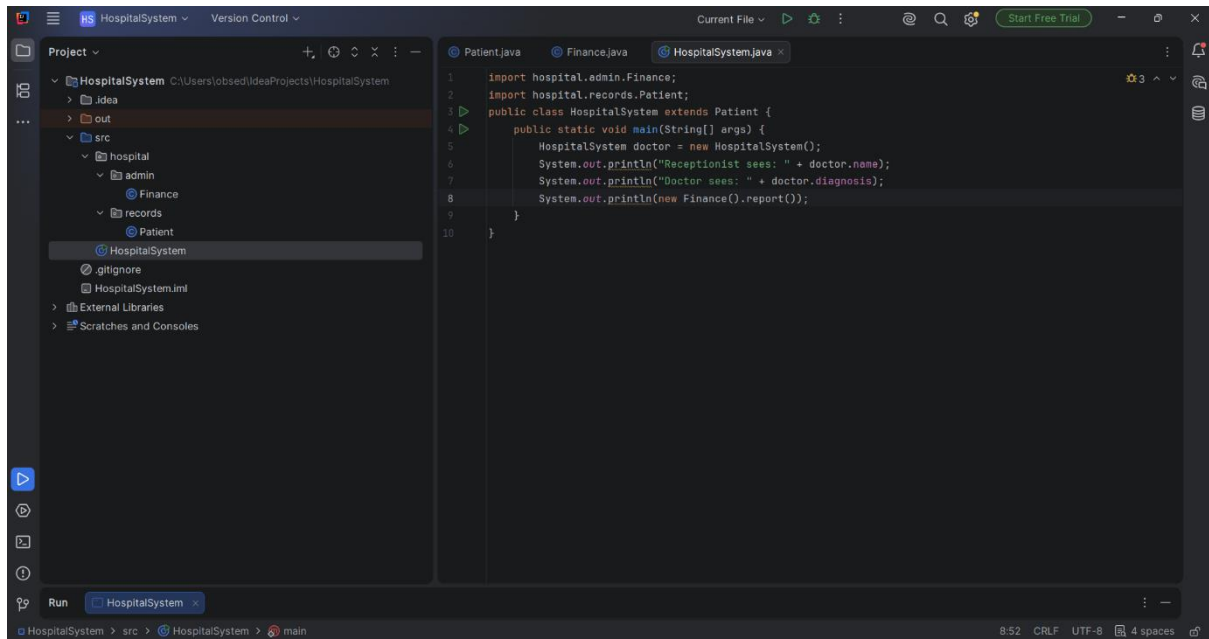
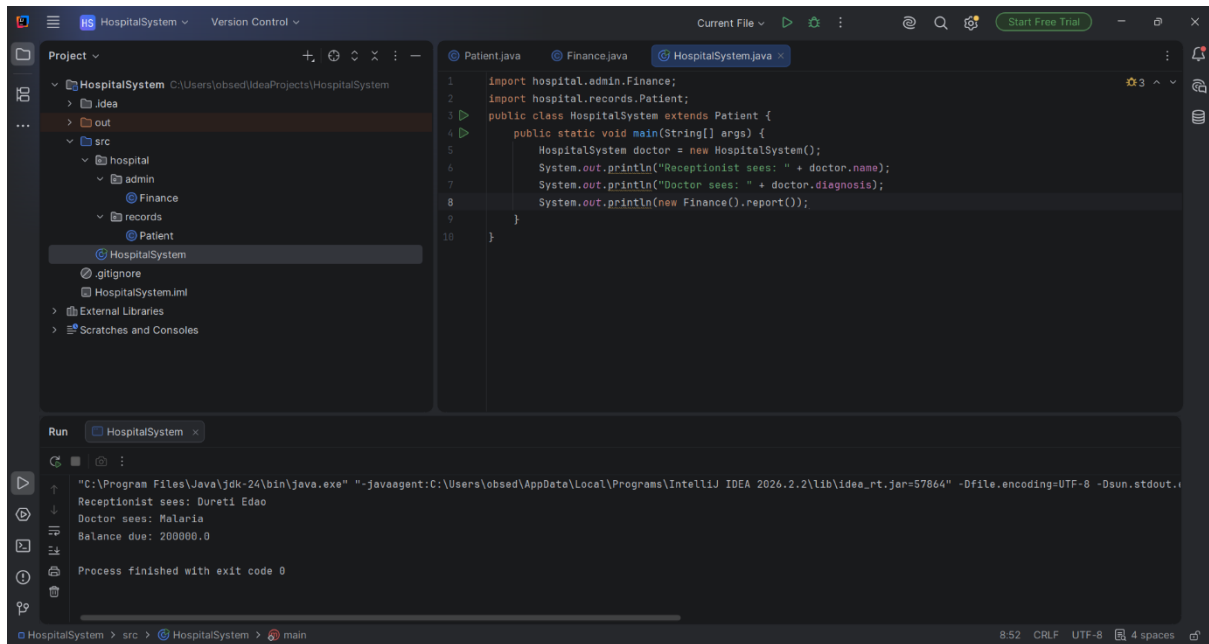


Figure 1:Class Main showing project structure

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**Output and caption descriptions**



```
1 import hospital.admin.Finance;
2 import hospital.records.Patient;
3 public class HospitalSystem extends Patient {
4     public static void main(String[] args) {
5         HospitalSystem doctor = new HospitalSystem();
6         System.out.println("Receptionist sees: " + doctor.name);
7         System.out.println("Doctor sees: " + doctor.diagnosis);
8         System.out.println(new Finance().report());
9     }
10 }
```

Run HospitalSystem

"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Users\obsed\AppData\Local\Programs\Intel\IJ IDEA 2026.2.2\lib\idea\_rt.jar=57864" -Dfile.encoding=UTF-8 -Dsun.stdout...

Receptionist sees: Dureti Edao  
Doctor sees: Malaria  
Balance due: 200000.0  
Process finished with exit code 0

Figure 2: Output after running the program

**Output Explanation**

When the program is executed, it displays the following four lines:

1. Receptionist sees: Jane Akello, 2026-10-05. The receptionist can view only the basic details, through public methods.
2. Doctor views diagnosis: Malaria. The doctor reads the medical information through the public method of MedicalFile, which uses the protected `getDiagnosis()`.
3. Doctor updated diagnosis: Malaria - treatment started. The doctor changes the private diagnosis through the protected `setDiagnosis()`, and the new value is shown.
4. Administration: Jane Akello balance due = 300000.0. Only the administration package can use the package-private `FinancialRecord` to work out the balance (500000 billed minus 200000 paid).

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**LOGIC BUILDING PROGRAMMING & VERSION CONTROL REPORT**

Conclusion: private hides the data, public gives controlled access, protected limits access to subclasses, and package-private keeps financial records inside the administration package. Trying to access hidden members from the wrong class causes a compile error.



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**LOGIC BUILDING PROGRAMMING & VERSION CONTROL REPORT**

**Section 2**  
**Version Control**

**Repository Overview**

|                        |  |
|------------------------|--|
| <b>Repository Name</b> |  |
| <b>Repository URL</b>  |  |
| <b>Branches</b>        |  |

**Commit #1 On Branch #1**

*[Insert your GitHub Desktop commit screenshot here]*

Figure 3: Commit 1 on Branch 1

**Commit Summary**

|                           |  |
|---------------------------|--|
| <b>Commit Name</b>        |  |
| <b>Commit ID</b>          |  |
| <b>Commit Description</b> |  |
| <b>Branch Affected</b>    |  |

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**LOGIC BUILDING PROGRAMMING & VERSION CONTROL REPORT**

**Section 3**

**What Have You Learned From The Task?**

| <b>Skill Set</b>                | <b>Description</b>                                                                                                                         |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Access modifiers                | Learned how private, protected, public and package-private (default) control who can see and change variables, methods and classes.        |
| Packages and inheritance        | Organised classes into packages (records, medical, admin) and used inheritance so a subclass in another package can use protected methods. |
| Data security and encapsulation | Understood how to protect sensitive data such as diagnoses and financial records by hiding it and exposing only what each role needs.      |